INCH-POUND

MIL-DTL-83197B
12 February 1996

SUPERSEDING
MIL-L-83197A(USAF)
6 December 1968

DETAIL SPECIFICATION

LINER, ANTI-EXPOSURE, FLYING COVERALLS CWU-23/P

1. SCOPE

1.1 Scope. This specification establishes the requirements for one type of CWU-23/P anti-exposure flying coveralls liner.

1.2 Classification. The liner shall be furnished in the following sizes, as specified (see 6.2):

Size 1 (32-36)	Size 5 (37-40)	Size 9 (41-44)
Size 2 (32-36)	Size 6 (37-40)	Size 10 (45-47)
Size 3 (32-36)	Size 7 (41-44)	Size 11 (45-47)
Size 4 (37-40)	Size 8 (41-44)	Size 12 (45-47)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, weather or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: San Antonio AFMC/TILDD, 306 Tinker Drive BLDG 207, Kelly AFB, TX 78241-5916 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8475

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SPECIFICATIONS

FEDERAL

V-F-106	-	Fastener, Slide, Interlocking
V-T-295	-	Thread, Nylon
DDD-B-1199	-	Binding Textile, Cotton, Bias-Cut
DDD-L-20	-	Label: for Clothing, Equipage, and Tentage, (General Use)

MILITARY

MIL-C-18387	-	Cloth, Twill, Cotton, Fire Retardant Treated
MIL-C-81393		Cloth, Knitted, Polyamide, High Temperature Resistant, Simplex, Jersey

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-601	-	Rubber: Sampling and Testing
FED-STD-751	-	Stitches, Seams, and Stitchings

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

AIR FORCE

68J240-Patterns, Liner, Anti-Exposure, Flying Coveralls CWU-23/P, Size 1
68J241-Patterns, Liner, Anti-Exposure, Flying Coveralls CWU-23/P, Size 2
68J242-Patterns, Liner, Anti-Exposure, Flying Coveralls CWU-23/P, Size 3
68J243-Patterns, Liner, Anti-Exposure, Flying Coveralls CWU-23/P, Size 4
68J244-Patterns, Liner, Anti-Exposure, Flying Coveralls CWU-23/P, Size 5
68J245-Patterns, Liner, Anti-Exposure, Flying Coveralls CWU-23/P, Size 6
68J246-Patterns, Liner, Anti-Exposure, Flying Coveralls CWU-23/P, Size 7
68J247-Patterns, Liner, Anti-Exposure, Flying Coveralls CWU-23/P, Size 8
68J248-Patterns, Liner, Anti-Exposure, Flying Coveralls CWU-23/P, Size 9
68J249-Patterns, Liner, Anti-Exposure, Flying Coveralls CWU-23/P, Size 10
68J250-Patterns, Liner, Anti-Exposure, Flying Coveralls CWU-23/P, Size 11
68J251-Patterns, Liner, Anti-Exposure, Flying Coveralls CWU-23/P, Size 12

(Application for copies of the above listed drawings should be addressed to: San Antonio AFMC/TILDD, 306 Tinker Drive BLDG 207, Kelly AFB TX 78241-5916.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are those of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to American Society for Quality Control, 611 East Wisconsin Avenue, Milwaukee WI 53202-4606).

2.4 Order of precedence. In the event of conflict between the text of this document and the references cited herein, The text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Preproduction. When required (see 4.2 and 6.2), this specification makes provisions for preproduction testing.

3.2 Materials and components. Materials and components that are not covered by specifications and drawings or are not specifically described herein shall be of the best quality, as lightweight as practicable, and entirely suitable for the purpose intended.

3.2.1 Outer shell. The fabric for the outer shell of the liner shall be rachel knit polypropylene netting conforming to Style 8846 (manufactured by the Fablock Mills, Incorporated, 140 Spring Street, Murray Hill, New Jersey), or equal. The color shall be natural or white.

3.2.2 Interlining. The fabric for the interlining of the liner shall conform to MIL-C-18387 and shall be sage green, USAF color shade No. 1509 (see 6.5).

3.2.3 Wrist stretch inserts. The fabric for the wrist inserts shall conform to MIL-C-81393 and shall be sage green, USAF color shade No. 1509 (see 6.5).

3.2.4 Binding tape. The tape for binding the sleeve ends, leg ends, and neck opening shall be either 1-3/8 inches or 1-1/4 inches wide and shall conform to type II, class 2 of DDD-B-1199. The color of the binding tape shall be sage green, USAF color shade No. 1509 (see 6.5).

3.2.5 Thread. The thread for assembling the liner and overedging shall conform to type I, class A, letter size B of V-T-295. The color of the thread shall be sage green, USAF color shade No 1509 (see 6.5). The thread shall have good colorfastness to laundering.

3.2.6 Slide fastener. The slide fastener for the front closure shall conform to type I, size LM or M, style 15, open top stop, bridge bottom stop, two sliders (mouth to mouth), brass chain, and with short tab pulls of V-F-106. The tapes shall be cotton and shall be USAF color shade No. 1509 (see 6.5). The length of the slide fastener for each size liner shall be determined from the patterns for each size liner. The bottom stop of the slide fastener, when installed in the liner, shall be 3/4 inch (plus or minus 1/8 inch) from the crotch seam, and the top stop shall be not more than 1/2 inch (plus or minus 1/8 inch) from the finished edge of the neck opening.

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3.3 Design. The liner shall be a one piece garment fabricated from two plies of different fabrics, single-row stitched together at the edges and equipped with stretch inserts at the wrists. The liner shall have a slide-fastener closure, at the front, extending from the crotch to the neckline.

3.4 Construction. The liner shall be constructed in accordance with table II using stitched seams. The manufacturer is not required to follow the exact sequence of operations as listed in table I.

3.4.1 Stitches, seams, and stitchings. The stitches, seams, and stitchings specified in table II shall be in accordance with FED-STD-751. Stitches per inch shall be as specified in table I. Each row of seam stitching shall be straight and parallel to the seam. Unless otherwise specified, a tolerance of plus or minus 1/32 inch will be allowed in the nominal distance of stitches from the edge on the joining seams of the liner. The thread tension shall be adjusted and maintained so that there shall be no loose stitching and the thread lock is imbedded in the center of the materials being sewn together. Thread breaks, skips, and runoffs shall be overstitched not less than 1/2 inch. The ends of stitchings not caught in other seams or stitchings shall be backstitched not less than 1/2 inch.

3.5 Patterns. The liner shall conform to the patterns referenced on Drawing 68J240, 68J242, 68J242, 68J243, 68J244, 68J245, 68J246, 68J247, 68J248, 68J249, 68J250, or 68J251, as applicable. The patterns furnished shall be used as a guide for processing the manufacturer's working patterns. The patterns shall not be altered without prior approval of the procuring activity.

3.6 Identification of product. The liner shall have an identification label conforming to type I, classes 1 and 2 of DDD-L-20. The size of the label shall be approximately 1-3/4 by 2-3/4 inches. The label shall have good colorfastness to laundering. The label shall contain the following information.

Liner, Anti-Exposure, Flying Coveralls CWU-23/P
MIL-DTL-83197B
Part or Identifying Number (PIN) 1/ (see 6.3)
Size No. 1/ (see 1.2)
Stock No. 1/
Manufacturer's Name 1/
Contract No. 1/
Date of Manufacture 1/

1/ The manufacture shall insert the applicable information.

3.7 Workmanship. The liners shall be uniform in quality and shall be free from irregularities or any defects that could adversely affect the performance, reliability, or durability of the liners.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Preproduction inspection

b. Quality conformance inspection

4.2 Preproduction inspection. The preproduction inspection shall consist of the visual examinations as specified in 4.3.2.1 and tested as specified in 4.4.

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4.2.1 Preproduction samples. The preproduction samples shall consist of three liners, each of a different size, and shall be representative of the construction, workmanship, components, and materials to be used during production. Before production is commenced, the samples shall be submitted to the procuring activity for examination as specified in 4.3.2.1 to determine that the liners meet the requirements of this specification. When a contractor is in continuous production of items covered by this specification from contract to contract or within a 12-month period, submission of preproduction samples on the new contract may be waived at the discretion of the procuring activity (see 6.2). Approval of the preproduction samples or waiver of the preproduction samples shall not preclude the requirements for quality conformance inspection.

4.3 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with ANSI/ASQC Z1.4.

4.3.1 Components and materials. Unless otherwise specified herein or in the applicable acquisition documents (see 6.2), components and materials shall be tested in accordance with all requirements of referenced specifications, drawings, and standards of this specification.

4.3.2 Examination of the end item. The end item shall be visually examined for defects as specified in 4.3.2.1. The sample unit for the examination shall be one liner. The inspection level for the visual examination shall be II of ANSI/ASQC Z1.4, and the acceptable quality level (AQL) shall be as specified in the contract or purchase order.

4.3.2.1 Visual examination. The liners shall be visually examined for defects as specified in table II.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department System Command. Packaging data retrieval is available from the managing Military Department or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The liner covered by this specification is intended for continuous wear by aircrew members during overwater flights and is to be worn under the Coveralls, Flying, Anti-exposure CWU-21/P and in conjunction with the Underwear, Double-knitted, Anti-exposure CWU-22/P and the Socks, Rubber SRC-25/P.

6.2 Acquisition requirements. Acquisition documents must specify the following:

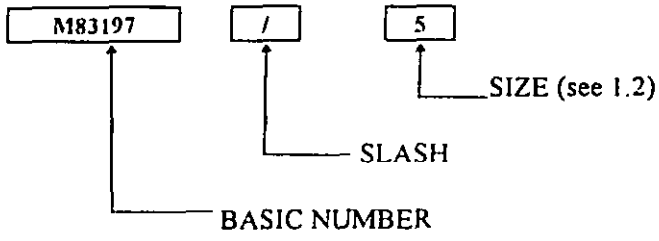
a. Title, number, and date of the specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2, and 2.3).
c. Part or Identifying Number (PIN) (see 6.3).
d. Waiver of preproduction samples if applicable (see 3.1 and 4.2.1).
e. Requirements for testing components and materials (see 4.3.1).
f. Packaging requirements (see 5.1).

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6.3 Part or Identifying Number (PIN). The PIN format is as follows:

Sample PIN - M83197/5

Construction of the PIN



6.4 Subject term (key word) listing.

Aircrew
Protective garment
Fowl weather
Under garment
Harsh environment

6.5 Color samples. Color samples may be obtained from the procuring activity or as directed by the contraction officer.

6.6 Figure. Figure 4 shows general design of the liner and is furnished as information only.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

TABLE I. Sewing operations.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
1	The liner shall be fabricated from the interlining cotton fabric specified in 3.2.2 and the polypropylene netting specified in 3.2.1. The cotton fabric be used as a carrier for the netting. The combined netting and cotton shall be used for the body and the sleeves or the liner. The cotton interlining shall be to the inside and the netting shall be to the outside.			
2	Join the fronts, the backs, and the sleeve parts of the cotton fabric and the netting together at all edges with	301	SSa-1	8-10

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TABLE I. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
	a single row of stitching not less than 1/8 inch nor more than 1/4 inch from the edges. edge.			
3	Butt join the slide fastener tape extensions at the bottom stop with a zigzag or two bartack rows of stitching across the tape widths. The zigzag stitching shall extend the length of the extension. One bartack shall be not less than 1/4 inch from the stop. The other bartack shall be equidistant from the first bartack and the end of the tape. the tape.	301	FSd-3(a) or bartack	10-12 20-24
4	Install the slide fastener to the two fronts in accordance with figure 1.			
5	Join the two fronts at the crotch seam, 5/16- or 1/4-inch gage.	401	LSc-2	8-10
6	Join the right and left back panels at the center back, right overlapping left, 5/16- or 1/4-inch gage.	401	LSc-2	8-10
7	Join the shoulder seams with a double-lap seam, with the fronts over the backs, 5/16- or 1/4 inch gage.	401	LSc-2	8-10
8	Join the topsleeve to the undersleeve at the back seam with a double lapped seam, 5/16- or 1/4-inch gage.	401	LSc-1	8-10
9	Join the sleeves to the body with a double-lapped seam, liner overlapping sleeves, 5/16- or 1/4-inch gage.	401	LSc-2	8-10
10	Join the side seams with a double-lapped seam, fronts overlapping backs, 5/16- or 1/4-inch gage.	401	LSc-2	8-10
11	Join inseam with a double-lapped seam, fronts over backs, 5/16-or 1/4-inch gage. The back seam and front crotch	401	LSc-2	8-10

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TABLE I. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
	seam shall match.			
12	Bind end of sleeve with bias tape, from cutout on one side to cutout on opposite side, using one row of stitching 1/16 inch from the turned-under edge. a. Trim off ends of binding even with sides of cutout. b. Turn sleeves and raise on netting side with a double row of stitching, 1/4-inch gage.	301	BSc-1	8-10
13	Set in knitted stretch fabric inserts at wrist in accordance with figure 2. The bottom folded edge (widest part) shall be even with bound end of the sleeve.	301 501 or 502	LSb-2 SSa-1	8-10 8-10
14	Stitch around the leg cutout, as shown on figure 3, with one row of stitching 1/4 inch (plus or minus 1/16 inch) from the cutout line marked from the pattern.	301	SSa-1	8-10
15	Make cutout and overedge stitch the raw edges of the netting and the cotton fabric together, covering the stitch row installed as operation 14.	501, 502, or 504	EFd-1	8-10
16	Bind the neck opening and the ends of of the legs with bias tape, using one row of stitching 1/16 inch from the turned-under edges. The tape shall continue around the full neck opening and around the leg endings from the cutout on one side to the cutout on the opposite side.	301	BSc-1	8-10
17	Turn under the raw ends of the tape 1/4 inch on the leg endings and neck opening to the inside liner, and stitch down 3/16 inch (plus of minus 1/16 inch) from the raw edge.	301	SSa-1	8-10
18	Install the label at the center back on the inside liner, 2 inches (plus or minus 1/8 inch) down from the	301	SSa-1	8-10

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TABLE I. Sewing operations - Continued.

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch
	neckline, with a single row of stitching on all sides 1/8 inch (plus or minus 1/16 inch) from the edges.			
19	Remove all loose threads, and trim the thread ends.			

TABLE II. Classification of defects.

Defect	Major	Minor	
		A	B
General			
Hole, scissors or knife cut, tear, burn, abraded area, oil or soil marks, or any other weakening defects resulting from the manufacturing process	X		
Color of components not as specified		X	
Thread ends not trimmed or loose thread ends not removed			X
Any spot or stain, such as solvents, that cannot be removed without damage to the liner fabric or components	X		
Any shade identification stamping visible on the outer side		X	
Any corrosive metal fastening device or sewn-on ticket	X		
Slide Fastener 1/			
Any part of a slide fastener bent, broken, or otherwise defective	X		
Length of slide fastener not as specified		X	
Component and Assembly			
Any component not as specified or any defect of a component or defect of assembly not classified herein 2/			

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TABLE II. Classification of Defects - Continued.

Defect	Major	Minor	
		A	B
General			
Any component, component part, or required operation omitted or any other operation improperly performed not specifically classified herein 2/			
Cutting			
Any component part not cut in accordance with patterns	X		
Seams and stitching			
Any seam twisted, puckered, or pleated (unless otherwise classified herein)		X	
Outer side of any sleeve or leg puckered, twisted, or pleated by the stitching joining the sleeve cuff or forming the leg hem		X	
Any cuff or hem twisted, puckered, or pleated affecting the smoothness of the outer surface of the sleeve leg		X	
Any part of the liner caught in a unrelated operation or stitching		X	
Any edge not properly forced out, i.e., having a fold 1/8 inch or more			X
Ends of stitching that are not caught in other seams or stitchings not securely tacked or backstitched	X		
Ends of seams or stitches that are not caught in other seams or stitchings secured but backstitched for a distance less than 1/2 inch		X	
Backstitching not overlapped on itself on ends of stitching that are not caught in other seams or stitching			X
Thread breaks not securely overstitched	X		
Thread breaks overstitched for a distance less than 1/2 inch		X	

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TABLE II. Classification of defects - Continued.

Defect	Major	Minor	
		A	B
General			
Ends of any armhole seam, at the side seam, or ends of any leg inseam, at the crotch, staggered by 1/2 inch or more		X	
Gage of stitching irregular, i.e., unevenly gauged or various stitching not uniform (to be scored only when the condition exists on the major portion of the seam)		X	
Gage of stitching beyond the range of width specified			X
Open seam except on hem, up to and including 1/4 inch 3/			X
Open seam except on hem, more than 1/4 inch but not more than 1 inch 3/		X	
Open seam, except on hem, more than 1 inch 3/	X		
Open seam, on any hem, 1/2 inch of less 3/			X
Open seam, on any hem, more than 1/2 inch 3/		X	
Raw edges on outer side of liner 4/			
-more than 1/8 inch but not more than 1/4 inch			X
-more than 1/4 inch but not more than 1 inch		X	
-more than 1 inch	X		
Raw edges on inner side of liner 4/			
-more than 1/4 inch but not more than 1 inch			X
-more than 1 inch		X	
Not specified seam or stitch type	X		
Looper thread on the outer side on liner when stitch type 401 is used	X		
Stitch tension loose resulting in a loose seam	X		
Stitch tension tight resulting in breakage of stitches when normal pull is applied 5/	X		
Stitches per inch one or two stitches more or less than specified, except on labels 6/			X
Stitches per inch three more or less than specified, except on labels 6/		X	

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TABLE II. Classification of defects - Continued.

Defect	Major	Minor	
		A	B
General			
Stitches per inch more than three stitches more or less than specified, except on labels 6/	X		
Stitches per inch one to four stitches more or less than specified on labels 6/			X
Stitches per inch more than four stitches more or less than specified on labels 6/		X	
Label			
Improperly printed (not containing required information, mistake in information, etc.)	X		
Blurred printing (if illegible	X		
Faint but legible		X	
Printing improperly placed (too near top or bottom)			X

NOTES (TABLE II)

1/ Each slide fastener shall be checked for proper functioning by opening and closing it at least three times along its full length.

2/ If the defect seriously affects appearance or serviceability, it shall be classified as a major defect. If the defect does not seriously affect appearance or serviceability, it shall be classified as a minor A defect.

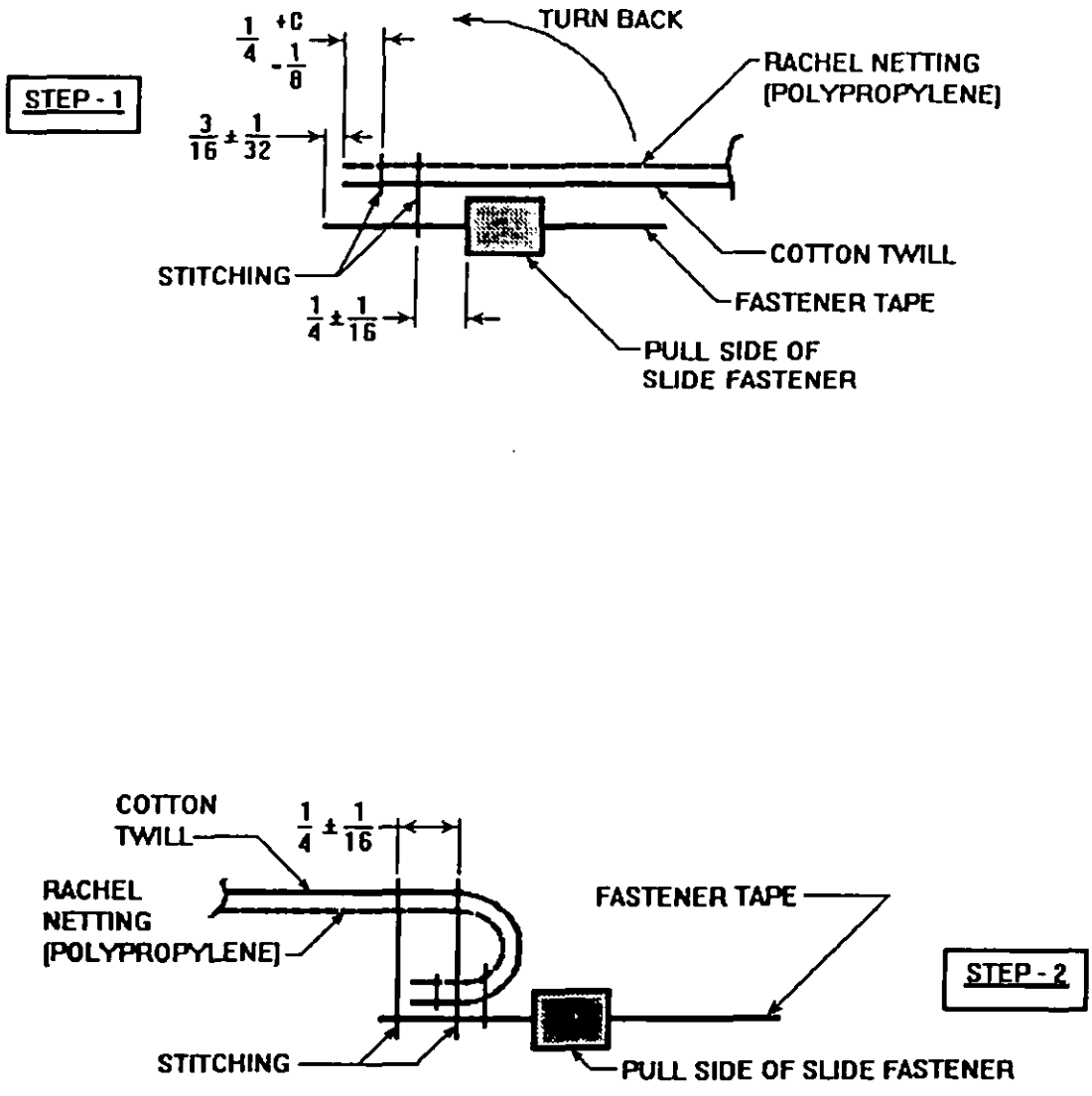
3/ If one or more stitches joining a seam are broken or if two or more continuous skipped stitches or runoffs occur, a seam shall be classified as open. If either one or both sides of a double-stitched seam is open, it shall be classified as an open seam.

4/ Any edge that is required to be turned under and is not but is securely caught in the stitching shall be classified as raw edge. Any raw edge not securely caught in the stitching shall be classified as an open seam.

5/ Puckering is evidence of tight tension. If puckering is evident, the seam shall be tested by exerting normal pull in the lengthwise direction of the seam or stitching.

6/ Stitches per inch shall be classified as a defect only when the condition exists on the major portion of the seam or label.

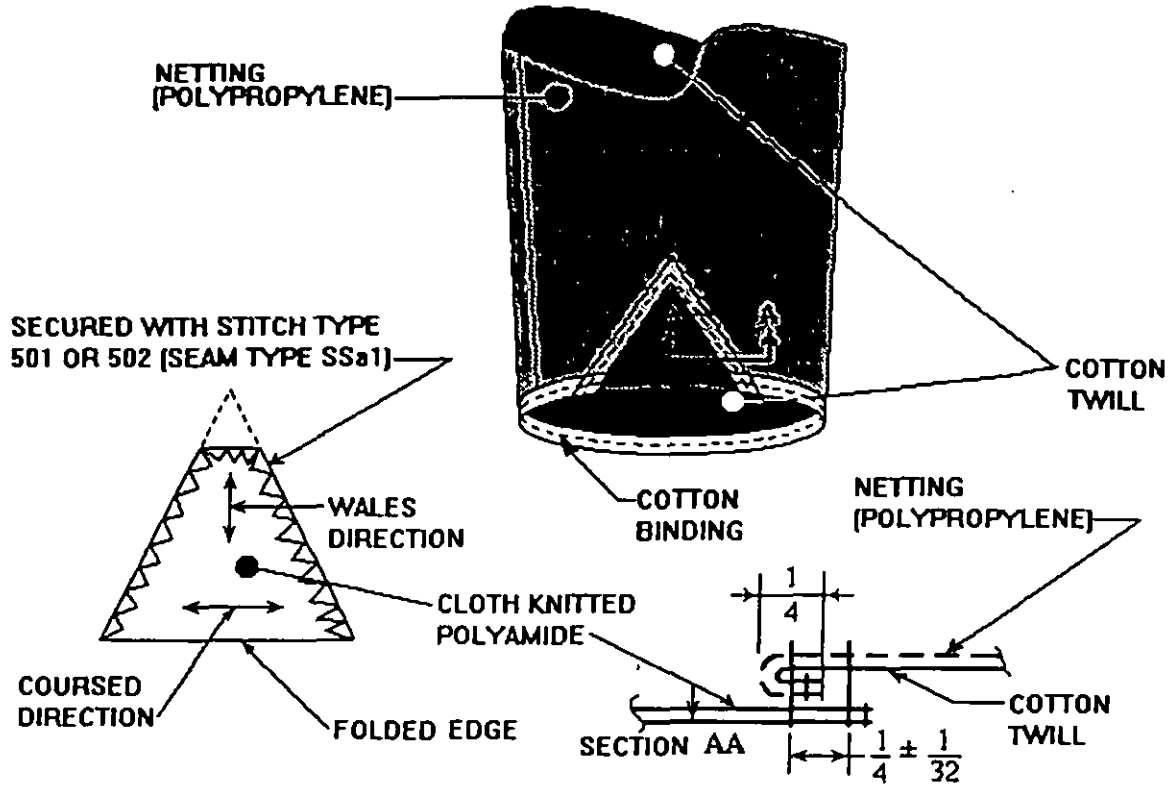
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NOTE: DIMENSIONS ARE IN INCHES

FIGURE 1. *Slide fastener installation (Right Front shown).*

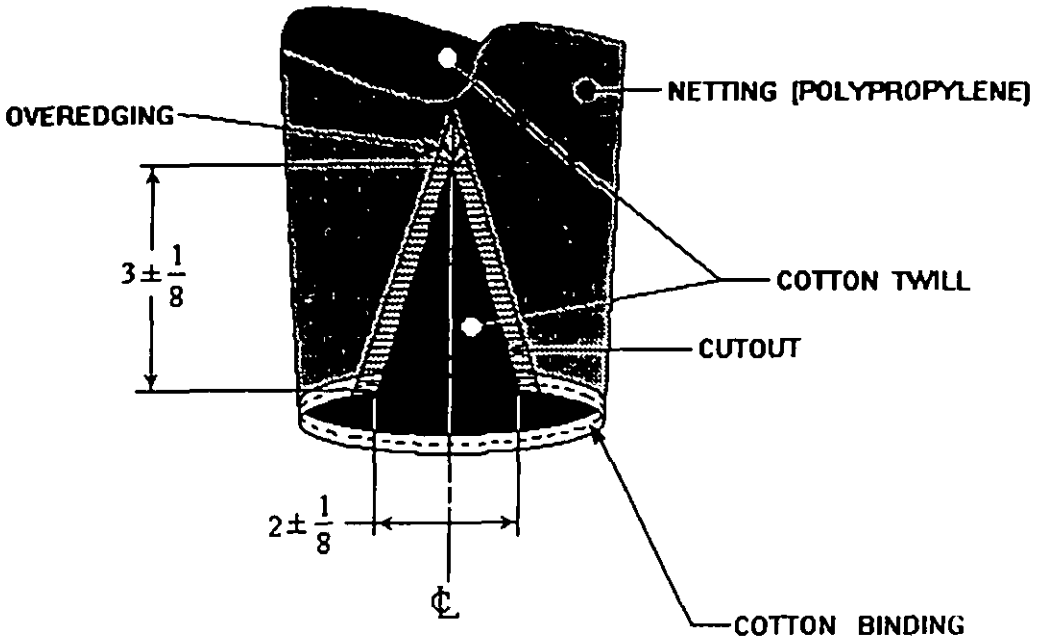
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NOTE: DIMENSIONS ARE IN INCHES

FIGURE 2. *Sleeve insert installation.*

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NOTES: 1. DIMENSIONS ARE IN INCHES
2. SHOWN, FRONT OF LEGG

FIGURE 3. *Leg cutout.*

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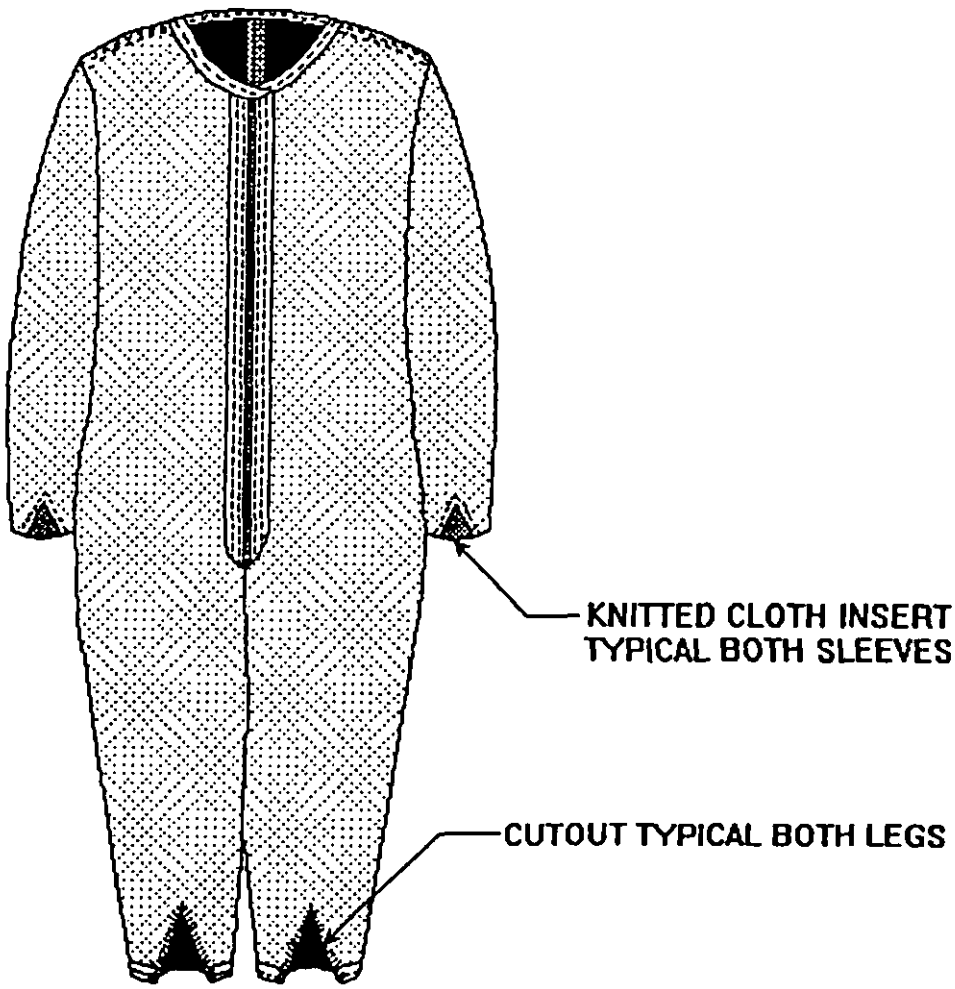


FIGURE 4. *Liner (Front View).*

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Custodians:
Army - GL
Navy - AS
Air Force - 99

Preparing activity:
Air Force - 82

Agent:
Air Force - 99

(Project 8475-F233)

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